



BESS Operational Availability Deep Dive

How to measure and compute if your BESS is meeting setpoint "Operational Availability"

Module 1 - BESS Availability Explained

Target Setpoint
Actual MW O

TRACKING SU

 by Fred Rivollier

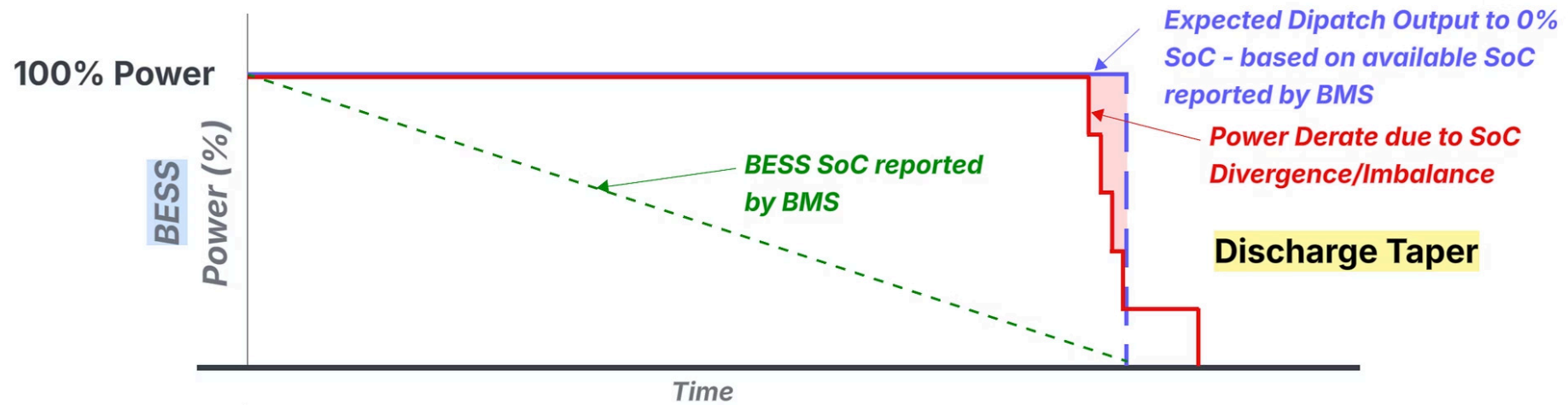
BESS Availability Types Recap

Combining all 4 availability type will form **OBE (Overall Battery Effectiveness)**

<u>Availability Type</u>	<u>Granularity</u>	<u>Coverage</u>	<u>Accuracy</u>	<u>Blind Spots</u>	<u>Primary Unit</u>	<u>Secondary Unit</u>
① Equipment Availability	MEDIUM (ON/OFF)	HIGH 100% of Calendar Time	HIGH (Equipment Status signals)	HIGH Imbalance, SoC error, Derates	%	MW_installed MW_contracted
② Operational Availability	HIGH (MW)	LOW Only During Dispatch	HIGH (Meter Measurement/ Setpoint)	HIGH All time with dispatch command < nameplate power	MW	%_contracted
③ Power Availability	HIGH (MW)	HIGH 100% of Calendar Time	MEDIUM Calculated by EMS. Based on various equipment reported metrics	MEDIUM Many corner case and special condition that might be hard to model.	MW	%_installed %_contracted
④ Energy Availability	HIGH (MWh)	MEDIUM 100% of Calendar Time but might only be updated with capacity tests annually	LOW to MEDIUM Calculated by EMS, include SoC Error and Imbalance	MEDIUM Might include power derated energy due to imbalance	MWh	%_installed %_contracted

② Operational Availability

<u>Granularity</u>	<u>Coverage</u>	<u>Accuracy</u>	<u>Blind Spots</u>	<u>Primary Unit</u>	<u>Secondary Unit</u>
HIGH (MW)	LOW Only During Dispatch	HIGH (Meter Measurement/ Setpoint)	HIGH All time with dispatch command < nameplate power	MW	%_contracted



example shortfall root-cause - SoC Averaging Effect

One Number Can't Tell the Whole Story

A single number
can't capture a
distributed
phenomenon

BESS
Average SoC

81.6%

DC Bus / PCS
Average SoC

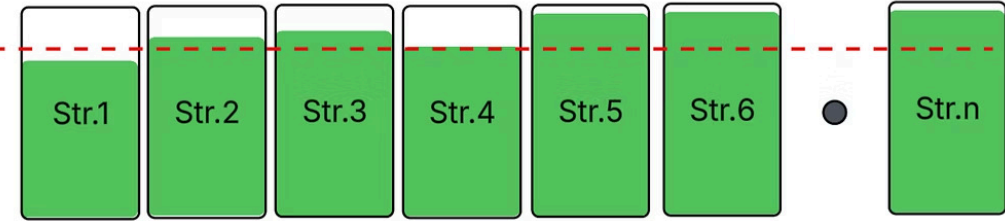
80%

90%

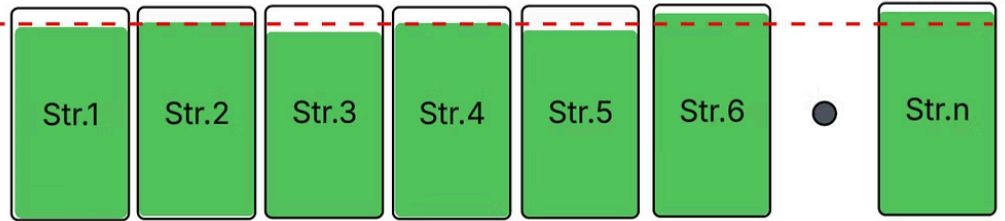
75%

*SoC Averaging Loses
Distribution Information*

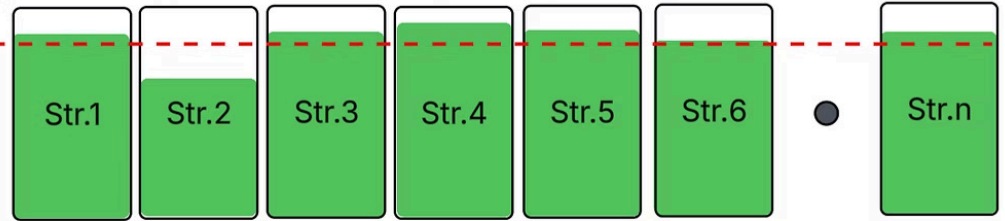
PCS 1



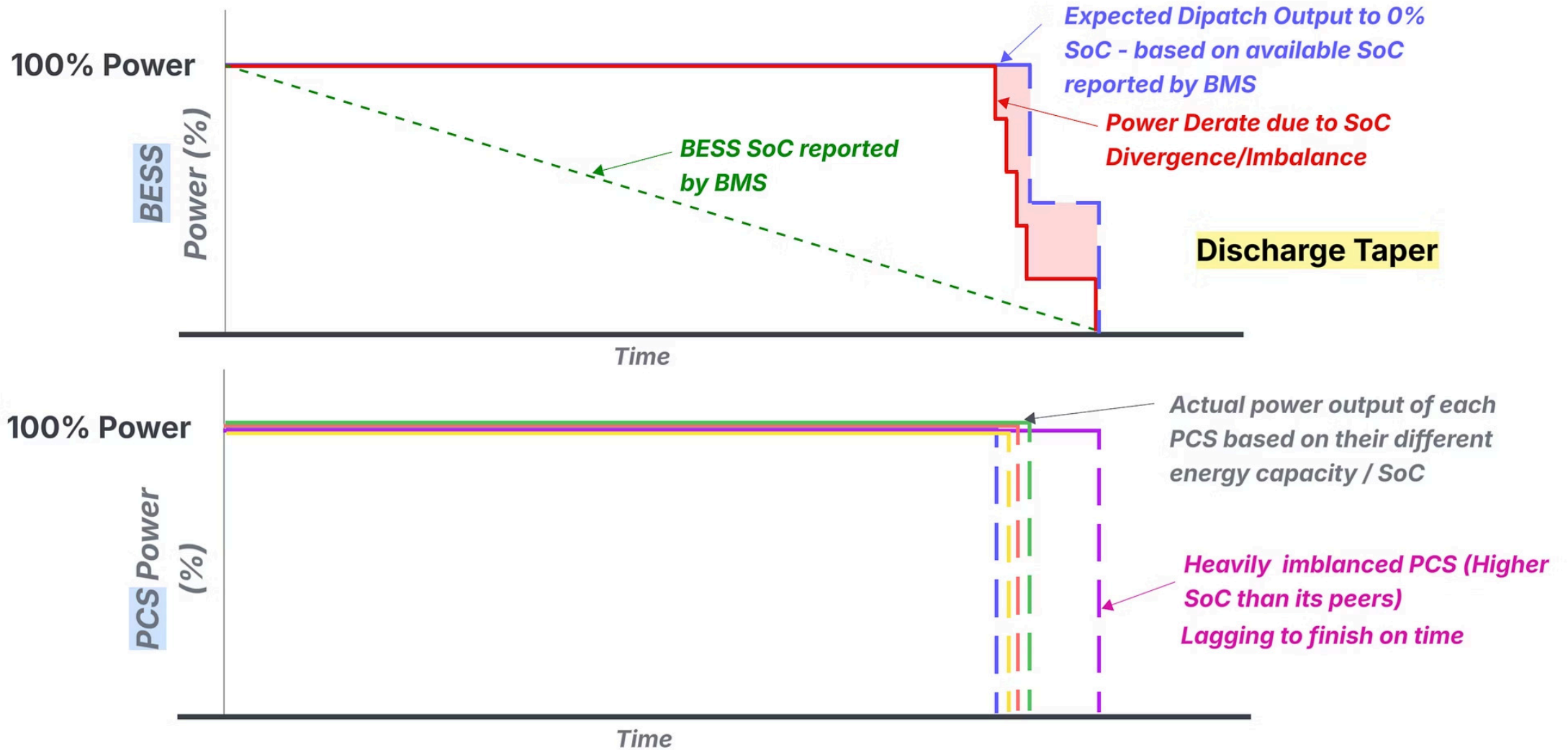
PCS 2



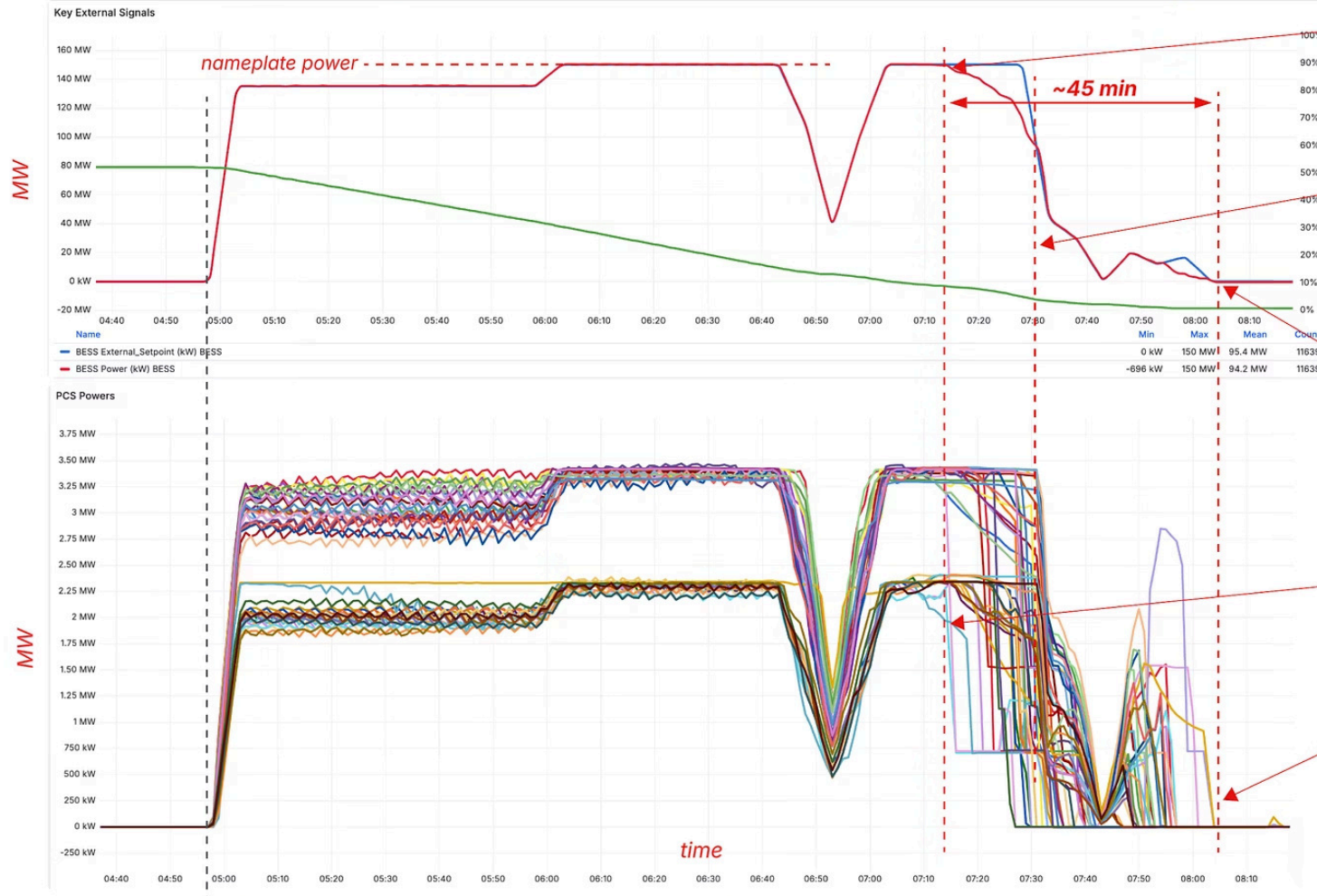
PCS 3



example shortfall - BESS End of Discharge Profile



example 4 hour BESS - 54 PCS



2026-03-26 07:14:08

BESS External_Setpoint (kW) BESS	150 MW
BESS Power (kW) BESS	150 MW
BESS SoC (%) BESS	8.72%

derate starting 8.7% reported BESS SoC

2026-03-26 07:30:38

BESS External_Setpoint (kW) BESS	86.7 MW
BESS Power (kW) BESS	91.6 MW
BESS SoC (%) BESS	3.75%

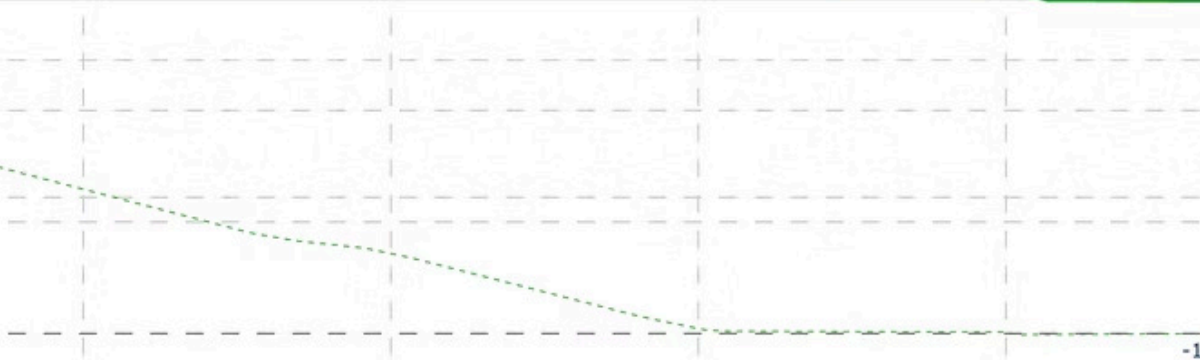
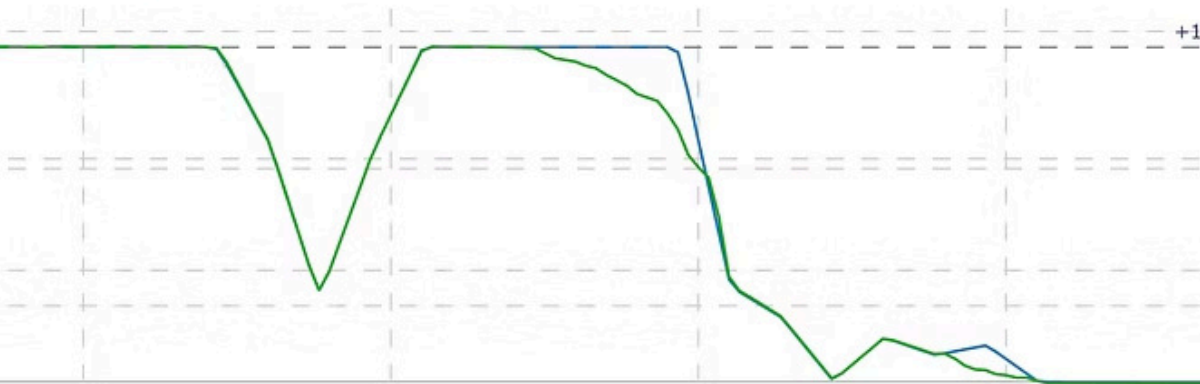
~ 15 min -- SoC ~ 4%

2026-03-26 08:03:33

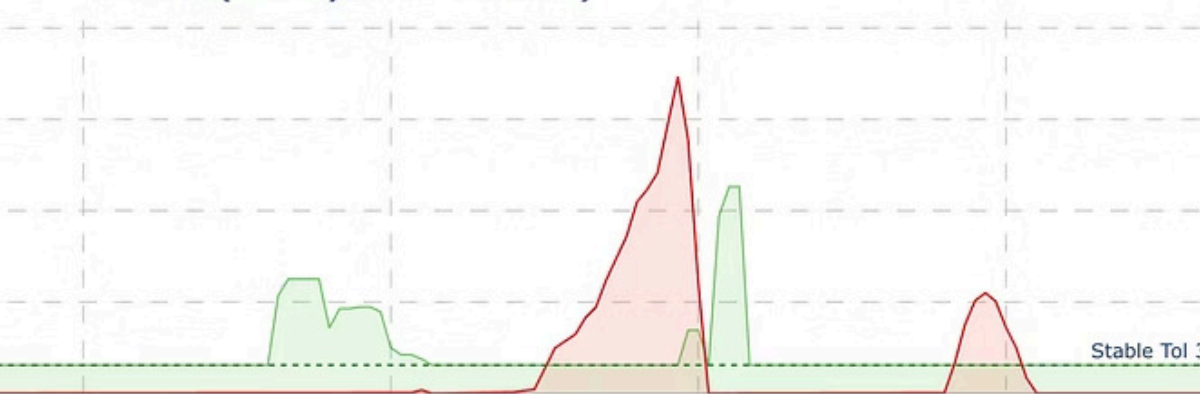
BESS External_Setpoint (kW) BESS	0 kW
BESS Power (kW) BESS	-681 kW
BESS SoC (%) BESS	0.658%

End of Discharge ~ 45 min from start of derate

Setpoint vs Actual Power



Shortfall (with Dynamic Tolerance)



Ramp Rate (%/sec of Nameplate)

example Shortfall Calculation

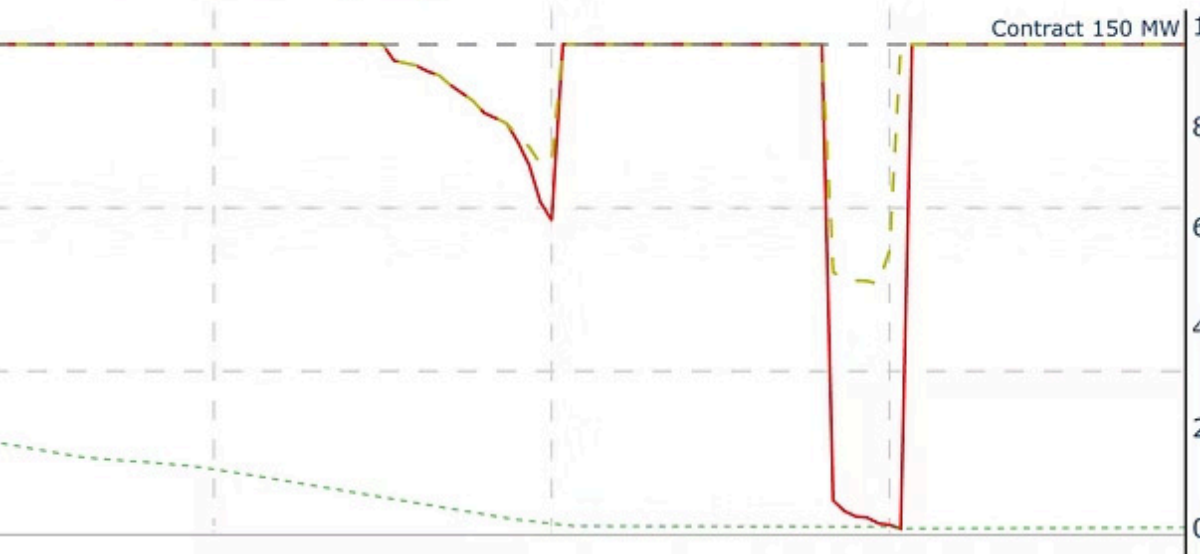
How to Compute Operating Shortfall (Set-point vs Actual)

Shortfall = $\max(0, |\text{setpoint}| - |\text{actual}|)$, absolute under-delivery only; over-delivery is always zero

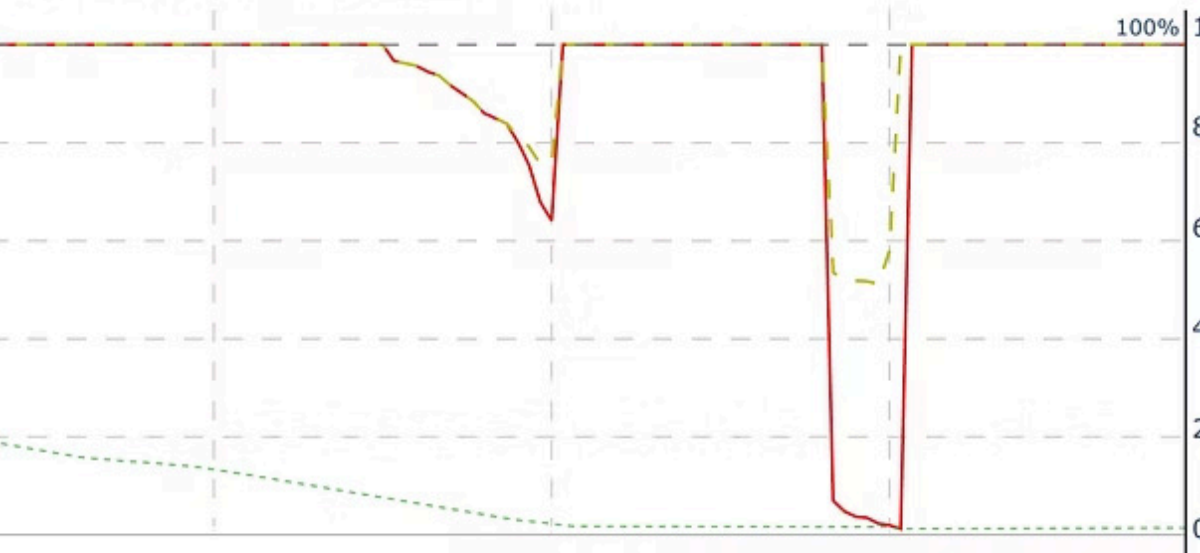
Dead-band = configurable tolerance (MW), shortfall within dead-band is not penalized to account for control accuracy.

The Dead-band can be proportionally increased by **Ramp-Rate** measured in the previous 2 intervals rolling max of **$\min(|\Delta\text{set-point}|, |\Delta\text{actual}|)$** as % of Contract MW/sec.

onal Available Power (MW)



Available Power (% of Contract)



Mar 26 07:00

Mar 26 07:30

Mar 26 08:00

Time

example Availability Calculation

How to Compute Operating Availability (Setpoint vs Actual)

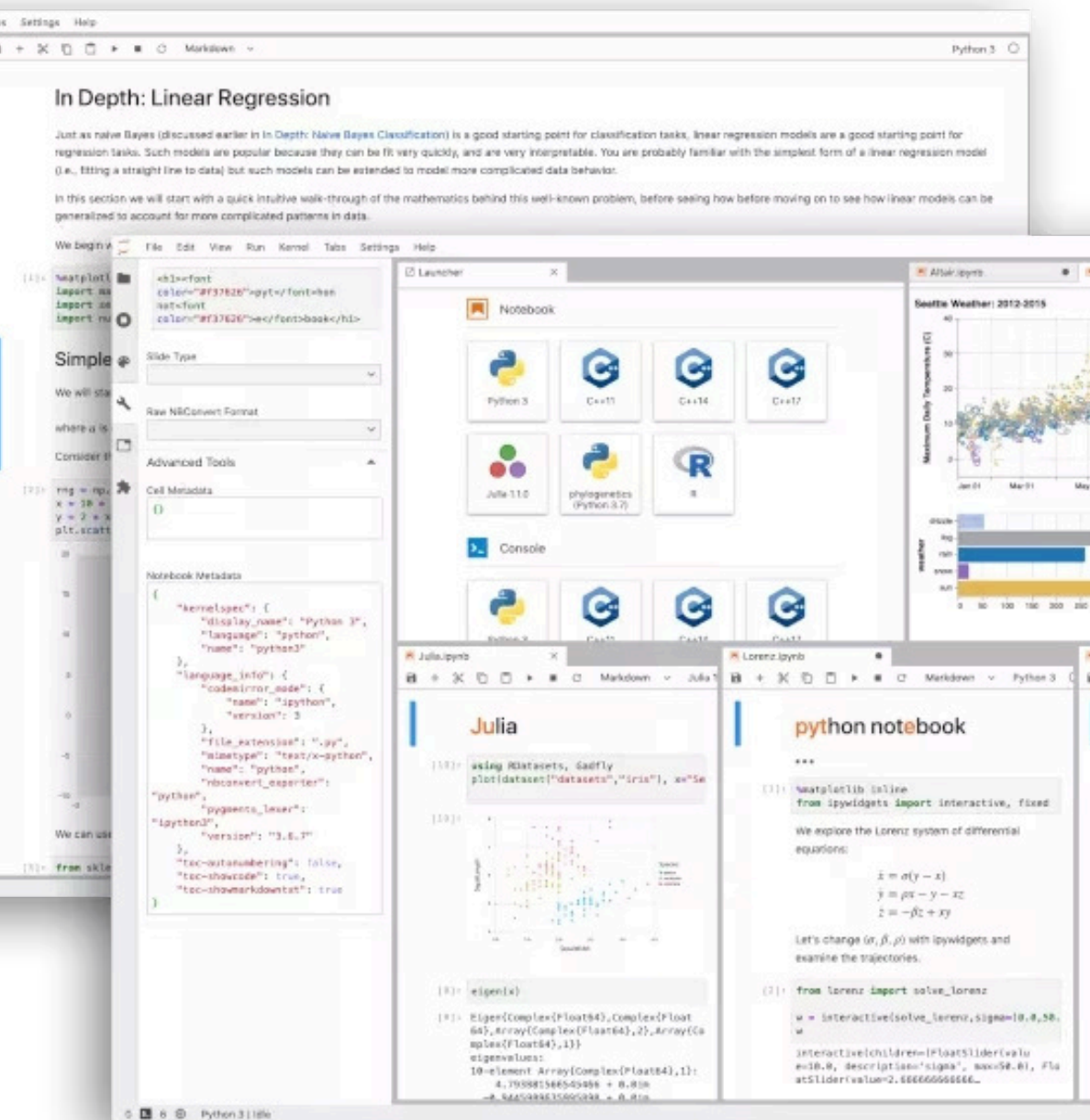
For every interval where the shortfall exceeds the dead-band **and** the SoC limit is not the cause, the actual output is the BESS available power.

For example, if 100 MW is requested and only 90 MW is delivered, the BESS is deemed 90 MW available.

The SoC exclusion is **directional**: a shortfall is only excused when the battery has hit its physical limit in the same direction as the request:

- SOC = 100% and set-point is **charging** → 100% available
- SOC = 0% and set-point is **discharging** → 100% available

If the directions oppose; an empty battery asked to charge, or a full battery asked to discharge; any shortfall counts fully against availability.



Processing with Python Notebooks

Dive into the practical application of BESS Operational Availability calculations using a **Python notebook**.

This interactive environment, such as **Marimo** or Jupyter, is ideal for performing complex calculations, generating insightful charts, and meticulously tracking every step of the process.

Let's explore an example to see these calculations in action.

What's Coming Next

We've covered the **Operational Availability**.

In our next sessions, we'll dive deeper into **Power Availability**.

Stay tuned for more Performance Engineering insights!

